

Technology Stack

Implement Client Script & UI Policy (Incident)

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Project Name	Implement Client Script & UI Policy (Incident)	Platform	ServiceNow

1. Technology Overview

The project is implemented within the ServiceNow platform and focuses on client-side configuration of the Incident table. The solution uses ServiceNow UI Policies, UI Policy Actions, and Client Scripts to control field behavior, automate values, validate submissions, and restrict direct list editing.

2. Technology Stack

Technology / Component	Purpose in the Project	Project Usage
ServiceNow	Enterprise service management platform used to build and configure the Incident solution.	Primary development and runtime platform.
Incident Table	Stores and manages Incident records and their fields.	Target table for all UI Policies and Client Scripts.
UI Policy	Applies conditional form behavior based on field values.	High Impact Control: triggered when Impact = 1 – High.
UI Policy Action	Defines the field-level behavior applied by a UI Policy.	Assignment group is mandatory; Urgency becomes read-only for High Impact.
Client Script – onChange	Runs when a monitored form field changes.	Sets Urgency to High when Impact changes to High.
Client Script – onSubmit	Validates data before a form is submitted.	Prevents saving a High Impact Incident when Assigned To is empty.
Client Script – onCellEdit	Controls direct editing of records from list views.	Prevents direct State changes through Incident list editing.
JavaScript	Scripting language used for client-side ServiceNow logic.	Used in the onChange, onSubmit, and onCellEdit Client Scripts.
ServiceNow Form / List UI	User interface through which Incident records are entered and edited.	Provides immediate field behavior, validation messages, and list-edit interaction.

3. Technology-to-Requirement Mapping

Requirement	Technology / Feature	Implementation
Conditional field control	UI Policy + UI Policy Action	High Impact condition makes Assignment group mandatory and Urgency read-only.
Automatic field update	onChange Client Script + JavaScript	Impact change to High automatically sets Urgency to High.
Save validation	onSubmit Client Script + JavaScript	Submission is stopped when Impact is High and Assigned To is empty.
List interaction restriction	onCellEdit Client Script + JavaScript	Direct State changes from the Incident list are blocked.
Immediate user feedback	ServiceNow form messaging / alert	Informational or error messages explain automatic updates and blocked actions.

4. Technology Architecture

Presentation / User Interaction Layer: ServiceNow Incident Form and Incident List provide the interface for entering and updating Incident data.

Client Control Layer: UI Policies and UI Policy Actions dynamically control field behavior. Client Scripts execute JavaScript for onChange, onSubmit, and onCellEdit events.

Data Layer: The ServiceNow Incident table stores valid Incident information after the configured client-side checks allow the update.

5. Development Environment

The implementation requires access to a ServiceNow instance with configuration permissions. The project configuration is created through ServiceNow administration interfaces such as UI Policies and Client Scripts. No separate frontend framework or external database is required for the implemented scope.

6. Technology Selection Rationale

ServiceNow native configuration components were selected because the project specifically targets the Incident form and requires immediate client-side control. UI Policies provide declarative field behavior, while Client Scripts provide event-based logic for automatic updates, validation, and list-edit restrictions. JavaScript supports the custom client-side conditions and actions required by the project.

7. Summary

The technology stack is centered on ServiceNow, the Incident table, UI Policies, UI Policy Actions, Client Scripts, and JavaScript. Together, these components implement the project's conditional field control, automatic value handling, save validation, and restricted list interaction without requiring an external application stack.